No limit cycle in the Banaji–Boros–Hofbauer three-reaction example

chemistry-open-problems · 18 October 2026

The question

Banaji, Boros and Hofbauer [1] classify the three-reaction, planar, quadratic mass-action systems that admit an Andronov–Hopf bifurcation (their Theorem 9, cases 9 and 10). In their Conclusions (§6, p. 22 of arXiv:2304.02303) they write:

> "However, we leave it open whether the networks in cases 9 and 10 admitting no Andronov–Hopf bifurcations can have a limit cycle. For example, we do not know whether the system $2\mathrm{X} \to 3\mathrm{X} + 2\mathrm{Y}$ (κ_1), $\mathrm{X} + \mathrm{Y} \to 0$ (κ_2), $\mathrm{Y} \to \mathrm{X} + \mathrm{Y}$ (κ_3), $\dot{x} = \kappa_1 \mathrm{x}^2 - \kappa_2 \mathrm{xy} + \kappa_3 \mathrm{y}$, $\dot{y} = 2\kappa_1 \mathrm{x}^2 - \kappa_2 \mathrm{xy}$ admits a limit cycle."

We settle the example (it has no periodic orbit at all, for any rate constants) and, with the same argument, the whole subfamily of case-9 networks to which it belongs.

Theorem 1. For all $\kappa_1, \kappa_2, \kappa_3 > 0$, the system

$$\dot{x} = \kappa_1 x^2 - \kappa_2 xy + \kappa_3 y, \qquad \dot{y} = 2\kappa_1 x^2 - \kappa_2 xy$$

has no periodic orbit in the closed positive quadrant. In particular it has no limit cycle.

Theorem 2. Let c_1, c_3, d_1 be positive integers and consider the networks

$$2X \xrightarrow{\kappa_1} (2 + c_1)X + d_1 Y, \qquad X + Y \xrightarrow{\kappa_2} 0, \qquad Y \xrightarrow{\kappa_3} (1 + c_3)X + Y,$$

i.e. case 9 of [1, Theorem 9] with reaction vectors $c = (c_1, -1, c_3)$, $d = (d_1, -1, 0)$. The mass-action system

$$\dot{x} = c_1 \kappa_1 x^2 - \kappa_2 xy + c_3 \kappa_3 y, \qquad \dot{y} = d_1 \kappa_1 x^2 - \kappa_2 xy$$

has a periodic orbit for some κ only if $c_1 < d_1 < 2c_1$, which is exactly the region where it undergoes an Andronov–Hopf bifurcation [1, Theorem 9]. For $d_1 \le c_1$ or $d_1 \ge 2c_1$ it has no periodic orbit for any $\kappa > 0$. The example of Theorem 1 is $c_1 = c_3 = 1$, $d_1 = 2$, on the boundary $d_1 = 2c_1$ of the Hopf region.

1 Proof

Bendixson–Dulac on the open quadrant. Write $F = (f, g)$ for the vector field of Theorem 2 and take the Dulac function $B(x, y) = y^{-b}$ on $\mathbb{R}^2_{>0}$, with

$$b = 1 + \frac{2\kappa_1 (d_1 - c_1)}{\kappa_2}\,.$$

A direct computation (script `certify_dulac.py`) gives

$$y^{\,b+1}\,\operatorname{div}(BF) = Q(x, y) := \frac{d_1 \kappa_1 \big(2(c_1 - d_1)\kappa_1 - \kappa_2\big)}{\kappa_2}\; x^2 + 2 d_1 \kappa_1\, xy - \kappa_2\, y^2,$$

a quadratic form whose y^2 coefficient is $-\kappa_2 < 0$ and whose discriminant is

$$\Delta = 4d_1\kappa_1(\kappa_1(2c_1 - d_1) - \kappa_2).$$

(The exponent b minimises Δ over all Dulac functions y^{-b}.) If $d_1 \geq 2c_1$, then $\kappa_1(2c_1 - d_1) - \kappa_2 \leq -\kappa_2 < 0$, so $\Delta < 0$ and Q is negative definite: $\operatorname{div}(BF) < 0$ on the open quadrant. Since $\mathbb{R}^2_{>0}$ is simply connected, the Bendixson–Dulac criterion excludes periodic orbits contained in it.

The boundary. On $\{y = 0, x > 0\}$ we have $\dot{y} = d_1\kappa_1 x^2 > 0$, and on $\{x = 0, y > 0\}$ we have $\dot{x} = c_3\kappa_3 y > 0$; the origin is an equilibrium. A periodic orbit in the closed quadrant cannot contain an equilibrium, and it cannot touch a boundary point where the field points strictly into the open quadrant (it would have to reach that point from inside). Hence every periodic orbit would lie in the open quadrant, which is impossible when $d_1 \geq 2c_1$.

The case $d_1 \leq c_1$. The stoichiometric matrix $\Gamma = \begin{pmatrix} c_1 & -1 & c_3 \\ d_1 & -1 & 0 \end{pmatrix}$ has rank 2 (its rows are not proportional since $c_3 \neq 0$), so $\ker\Gamma$ is spanned by $u = c \times d = (c_3,\ c_3d_1,\ d_1 - c_1)$. A positive equilibrium x requires a positive flux vector $v(x) \in \ker\Gamma \cap \mathbb{R}^3_{>0}$, i.e. u or $-u$ positive. If $d_1 \leq c_1$ this fails, so the system has no positive equilibrium. A periodic orbit in the open quadrant would enclose an equilibrium in its interior (index theory), which would be positive; hence there is none, and the boundary argument above excludes orbits touching the axes.

Theorem 1 is the case $c_1 = c_3 = 1$, $d_1 = 2$: $b = 1 + 2\kappa_1/\kappa_2$, $Q = -\frac{2\kappa_1(2\kappa_1+\kappa_2)}{\kappa_2}\,x^2 + 4\kappa_1 xy - \kappa_2 y^2$, $\Delta = -8\kappa_1\kappa_2 < 0$. For the record, the unique positive equilibrium is $(2\kappa_3/\kappa_2,\ 4\kappa_1\kappa_3/\kappa_2^2)$ with $\det J = 4\kappa_1\kappa_3^2/\kappa_2 > 0$ and $\operatorname{tr} J = -2\kappa_3 < 0$: it is globally the only recurrent object in the open quadrant. □

Why the example sits on the Hopf boundary. For case 9 with $c_3 > 0$, Theorem 9 of [1] gives an Andronov–Hopf bifurcation iff $\frac{1}{2}\,(d_3/c_3 + d_1/c_1) < d_2/c_2 < d_1/c_1$. In the subfamily, $d_2/c_2 = 1$ and $d_3 = 0$, so the condition reads $c_1 < d_1 < 2c_1$. For the example, $\frac{1}{2}\,(0 + 2) = 1 < 1 < 2$ fails by equality. Theorem 2 shows that in this subfamily, "no Andronov–Hopf bifurcation" implies "no periodic orbit".

2 What remains open

The general question of [1] (all case-9 and case-10 networks without an Andronov–Hopf bifurcation: case 9 with $d_2 \neq -1$ or $d_3 \neq 0$, including the branch $c_3 = 0$, and all of case 10) is not settled by this note. The same Dulac strategy with $B = x^\alpha y^\beta$ reduces it to sign conditions on a cubic-plus-quadratic form; this is being attacked separately.

3 Verification

`scripts/certify_dulac.py` checks every identity above symbolically (sympy): the Dulac computation, the coefficients and the discriminant in the general subfamily and in the example, the vector $u = c \times d$, the equilibrium and its Jacobian, and the boundary behaviour. Log: `certificates/certify_dulac.log`.

References

1. M. Banaji, B. Boros, J. Hofbauer, *Oscillations in three-reaction quadratic mass-action systems*, Stud. Appl. Math. 152 (2024) 249–278, doi:10.1111/sapm.12639, arXiv:2304.02303.